Pricing, Greeks, and Jump-Risk Management for BTC/ETH Vanilla Options

IEDA4331 Financial Engineering Group Project

Choi Man Hou
mhchoiaf@connect.ust.hk
SID: 20894196

Lee Cheuk Yiu
cyleebt@connect.ust.hk
SID: 21051040

Market data snapshot: 2026-05-12 14:51 UTC

Abstract

This report prices BTC and ETH vanilla options using the core IEDA4331 toolkit: risk-neutral valuation, Black-Scholes pricing, CRR binomial trees, implied volatility, Greeks, and dynamic hedging. We treat the Merton jump-diffusion model as a classical jump benchmark rather than the final innovation. The modern exploration layer then evaluates four crypto-relevant candidates at the same empirical level: stochastic volatility with correlated jumps, market-regime implied stochastic volatility, time-varying jump intensity, and GARCH option pricing. The calibrated Merton benchmark reduces mean absolute pricing error by 75.5% for BTC and 59.8% for ETH relative to a historical-volatility Black-Scholes baseline. The equal-level candidate comparison shows that MR-ISVM gives the best BTC fit, MR-ISVM gives the best ETH fit, and SVCJ produces the largest standardized crash-protection premium. The main conclusion is that crypto option markets price non-Gaussian jump and volatility-regime risk explicitly; Black-Scholes remains a necessary no-arbitrage benchmark, but it is not sufficient as a standalone crypto risk engine.

Contents

1 Introduction and Course Motivation 3

2 Data and Market Snapshot 3

3 Baseline Pricing Framework 4

3.1 Black-Scholes Benchmark 4

3.2 CRR Binomial Tree 4

4 From Merton to Modern Crypto-Specific Extensions 5

4.1 Merton as Classical Jump Benchmark 5

4.2 Candidate A: Stochastic Volatility with Correlated Jumps 6

4.3 Candidate B: Market-Regime Implied Stochastic Volatility 6

4.4 Candidate C: Time-Varying Jump Intensity 6

4.5 Candidate D: GARCH Option Pricing 7

5 Historical Distribution and Jump Diagnostics 9

6 Market Calibration and Statistical Evidence 10

7 Interest Rate Sensitivity 11

8 Jump-Parameter Sensitivity 12

9 Greeks and Risk Management 12

10 Jump-Hedging Simulation 13

11 Discussion 14

12 Conclusion 15

A Implementation Notes 16

B AI Assistance Statement 16

1 Introduction and Course Motivation

This report studies BTC and ETH vanilla options under the central language of IEDA4331: no-arbitrage pricing, risk-neutral valuation, binomial replication, the Black-Scholes-Merton PDE, implied volatility, Greeks, and hedging error. The objective is to move from classroom pricing identities to a market-calibrated crypto option risk engine, then ask where the classical diffusion model fails. We use the Black-Scholes model [2, 8], CRR binomial tree [4], and Merton jump-diffusion model [9] as the classical ladder of benchmarks.

The Risk-Neutral Pricing chapter brings the analysis from physical expected returns to martingale valuation. Instead of asking whether BTC or ETH is expected to appreciate, we price contingent claims under Q:

$$V_t = E_t^Q\left[e^{-r(T-t)}V_T\right].$$

The Binomial Tree chapter brings the analysis to finite-state replication: a derivative is priced by constructing a self-financing portfolio whose future payoff matches the claim. The Black-Scholes-Merton chapter then takes the limiting continuous-time view and derives the PDE from a self-financing delta hedge:

$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} - rV = 0.$$

The Risk Management chapter brings the analysis from valuation to control: Delta, Gamma, Vega, discrete hedging error, VaR logic, and stress testing. Our report follows that course arc, then adds a jump process precisely where the diffusion assumption breaks down for cryptocurrency markets.

2 Data and Market Snapshot

We use public Deribit data for BTC and ETH: `get_book_summary_by_currency` for option books, `get_instruments` for contract metadata, and `get_index_chart_data` for one-year index histories. Deribit option premiums are quoted in coin units, so we convert them into USD:

$$\text{Option price}_{USD} = \text{Option price}_{coin} \times S.$$

Table 1: Deribit option universe after cleaning

Currency	Raw rows	Usable rows	Median spot	Median IV	Min days	Max days
BTC	886	700	80655.1	42.38%	1.71	317.7
ETH	706	482	2272.8	57.67%	1.71	317.7

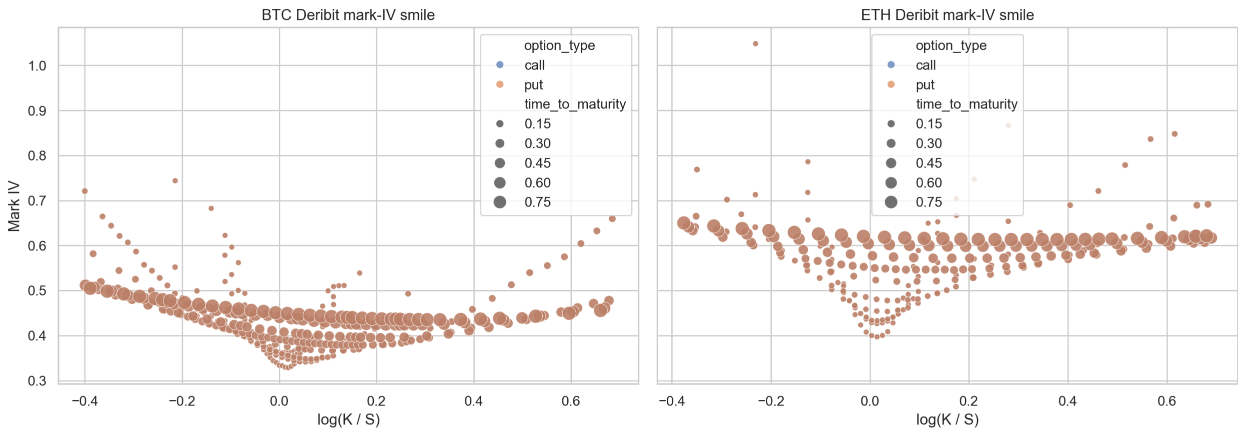

Figure 1: Deribit mark implied-volatility smiles across strikes and maturities.

Inference. A flat Black-Scholes volatility input would imply a roughly horizontal surface. The observed BTC and ETH smiles are curved and dispersed, indicating that the market prices skew, maturity effects, and non-Gaussian tail risk.

The course's implied-volatility concept is important here. Implied volatility is not simply another volatility estimator; it is the volatility that forces the Black-Scholes formula to match a traded option price. A smile therefore means the market is using Black-Scholes as a quote convention while rejecting its constant-volatility distributional assumption. Crypto option studies document precisely this pattern in Bitcoin options [12, 3]. This is the bridge from the data section to the model-innovation section.

3 Baseline Pricing Framework

3.1 Black-Scholes Benchmark

For European calls and puts with no dividend yield:

$$C = SN(d_1) - Ke^{-rT}N(d_2), \qquad P = Ke^{-rT}N(-d_2) - SN(-d_1),$$

where

$$d_1 = \frac{\log(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

Volatility is estimated from log returns, following the course treatment that log returns aggregate naturally across time and that annualized volatility scales with the square root of time. The physical drift is not used directly in risk-neutral pricing because the Black-Scholes PDE and risk-neutral valuation remove μ through replication.

3.2 CRR Binomial Tree

The CRR tree uses

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u}, \qquad p_Q = \frac{e^{r\Delta t} - d}{u - d}.$$

At each node:

$$V_t = e^{-r\Delta t}\big(p_Q V_u + (1 - p_Q)V_d\big).$$

For American options, the backward step replaces continuation value with

$$V_t = \max\{\text{continuation value},\ \text{immediate exercise value}\}.$$

This matters because the binomial tree is not only a numerical method. It is the discrete version of the course's no-arbitrage argument: p_Q is not the real-world probability that BTC or ETH goes up, but the probability that makes discounted prices consistent with replication.

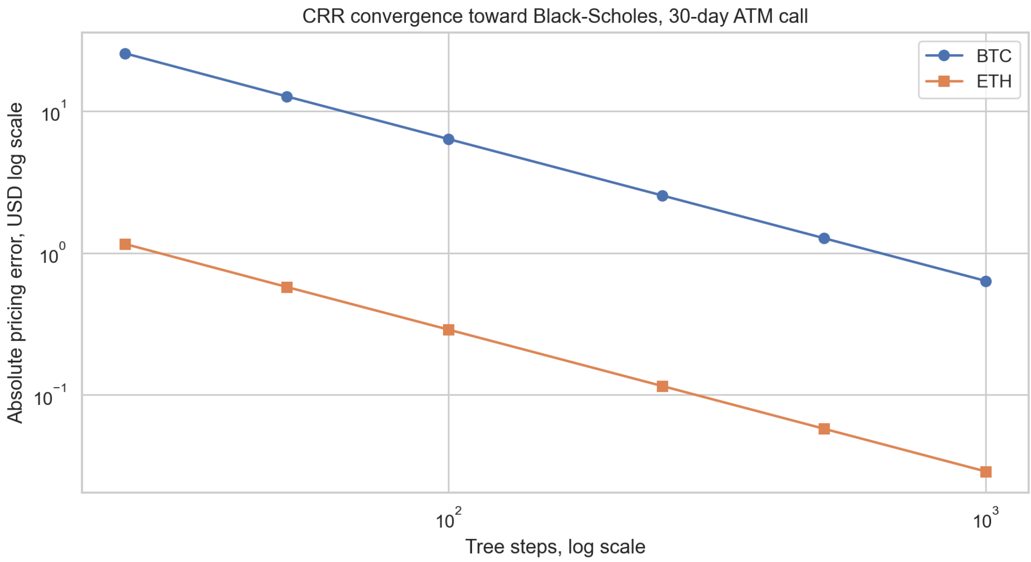

Figure 2: CRR convergence toward the Black-Scholes benchmark for 30-day at-the-money calls.

Table 2: European and American CRR pricing, 30-day at-the-money options

Currency	Option	Spot/Strike	BS Eur.	CRR Eur.	CRR Am.	Am. premium
BTC	call	80655.1	2574.10	2572.83	2572.83	0.00
BTC	put	80655.1	2507.88	2506.61	2510.01	3.40
ETH	call	2272.8	116.46	116.40	116.40	0.00
ETH	put	2272.8	114.60	114.54	114.63	0.09

Inference. The American call premium is effectively zero for non-dividend underlyings, while American puts carry a small early-exercise premium. This is exactly the qualitative result expected from the course theory: early exercise destroys call time value when there is no dividend benefit, but it can be rational for puts when the option is sufficiently in the money.

4 From Merton to Modern Crypto-Specific Extensions

4.1 Merton as Classical Jump Benchmark

The Merton jump-diffusion model was introduced in 1976 [9]. It is therefore not recent enough to be the final innovation by itself. In this report, it serves as the first serious extension beyond Black-Scholes: a classical benchmark that tests whether explicit price jumps improve crypto option pricing.

The model replaces the diffusion-only assumption with:

$$\frac{dS_t}{S_{t-}} = (r - \lambda\kappa)dt + \sigma dW_t^Q + (J-1)dN_t,$$

where

$$\log J \sim N(\mu_J, \delta_J^2), \qquad \kappa = E[J-1] = e^{\mu_J + \frac{1}{2}\delta_J^2} - 1.$$

Here λ is the annual jump intensity, μ_J is average log jump size, and δ_J is jump-size volatility. The compensator $\lambda\kappa$ keeps the discounted stock price consistent with risk-neutral valuation.

Conditional on n jumps, terminal log-price remains Gaussian. Hence the Merton option value is a Poisson mixture of conditional Black-Scholes-style prices:

$$V^{Merton}(S,K,T) = \sum_{n=0}^{\infty} e^{-\lambda T} \frac{(\lambda T)^n}{n!} V_n^{BS}(S,K,T).$$

This extension preserves the course's risk-neutral framework while adding a realistic channel for crypto discontinuities. The compensator is essential: without it, the jump term would change the drift of the discounted stock process and break the martingale logic that motivates risk-neutral pricing.

4.2 Candidate A: Stochastic Volatility with Correlated Jumps

The first modern candidate is a stochastic volatility with correlated jumps (SVCJ) model. This model family is closely related to affine jump-diffusion work in option pricing [1, 5], and it has been applied directly to cryptocurrency options in recent empirical work [6]. Its core idea is that crypto crashes are not only spot-price jumps; they are often volatility events as well.

A compact SVCJ specification is:

$$\frac{dS_t}{S_{t^-}} = (r - \lambda\mu_S)dt + \sqrt{v_t}\, dW_t^S + Y_t^S dN_t,$$

$$dv_t = \kappa_v(\theta_v - v_t)dt + \xi\sqrt{v_t}\, dW_t^v + Y_t^v dN_t, \qquad dW_t^S dW_t^v = \rho\, dt.$$

Here v_t is stochastic variance, ρ captures leverage/skew, and (Y_t^S, Y_t^v) allows price jumps and volatility jumps to arrive together. Compared with Merton, SVCJ can explain a more realistic crypto stress state:

$$\text{spot falls sharply} \quad + \quad \text{implied volatility jumps upward.}$$

4.3 Candidate B: Market-Regime Implied Stochastic Volatility

The second candidate is a market-regime implied stochastic volatility model (MR-ISVM), motivated by recent crypto option research [10]. The key idea is that crypto option surfaces behave differently across regimes such as calm markets, trending markets, and stress markets. A reduced formulation is:

$$z_t \in \{1, \dots, M\}, \qquad \sigma_{\mathrm{IV}}(K,T,t) = f_{z_t}(K,T),$$
$$V_t = BS(S_t, K, T, r, \sigma_{\mathrm{IV}}(K,T,t)).$$

The regime variable z_t can be estimated from returns, realized volatility, volume, or option-surface features. Compared with Merton, MR-ISVM does not force one set of jump parameters to explain all market states; it allows the implied volatility surface itself to become state-dependent.

4.4 Candidate C: Time-Varying Jump Intensity

The third candidate is a dynamic jump-intensity model. Recent Bitcoin option evidence emphasizes that jump magnitude and jump arrival rates are time-varying [3]. This is important for crypto because large moves can cluster through leverage, liquidations, and margin pressure.

A parsimonious extension is:

$$\frac{dS_t}{S_{t^-}} = (r - \lambda_t\kappa)dt + \sigma dW_t^Q + (J-1)dN_t,$$

$$d\lambda_t = \beta(\bar{\lambda} - \lambda_t)dt + \alpha\, dN_t.$$

After a jump, λ_t increases, then mean-reverts. Compared with Merton's constant λ, this model captures jump clustering and liquidation-cascade risk.

4.5 Candidate D: GARCH Option Pricing

The fourth candidate is a GARCH option-pricing model, which has also been studied for Bitcoin options [11]. It is a discrete-time alternative that connects naturally to the course's historical volatility estimation section:

$$r_{t+1} = \mu + \sqrt{h_{t+1}}\varepsilon_{t+1},$$
$$h_{t+1} = \omega + \alpha r_t^2 + \beta h_t.$$

Option prices can then be computed by simulation under a risk-neutralized conditional variance process. Compared with Merton, GARCH does not require discontinuous jumps, but it captures volatility clustering. It is therefore useful as a competing explanation for smiles: are crypto smiles mainly jump risk, or time-varying volatility?

Table 3: Equal-level implementation protocol for all candidate models

Model	Calibration object	Pricing engine	Same-level diagnostic
Black-Scholes	Historical log-return variance	Closed-form diffusion formula	Full-universe MAE/RMSE and Greeks
Merton	Weighted Deribit option-price error	Poisson mixture of Black-Scholes prices	Full-universe MAE/RMSE, jump sensitivity, hedging
SVCJ proxy	Historical variance state plus option variance multiplier	Moment-matched stochastic variance with Merton jumps	Full-universe MAE/RMSE and standardized crash put
MR-ISVM	Current-regime Deribit mark-IV cross section	Black-Scholes with fitted regime IV surface	Full-universe MAE/RMSE and standardized crash put
Dynamic jump	Weighted option-price error with mean-reverting λ_t	Merton mixture with horizon-dependent effective intensity	Full-universe MAE/RMSE and standardized crash put
GARCH	Historical conditional return variance	Black-Scholes with forecast average variance	Full-universe MAE/RMSE and standardized crash put

This protocol is deliberately symmetric. Each candidate uses the same cleaned Deribit universe, the same 80-contract calibration discipline where option calibration is needed, and the same out-of-sample-style diagnostic on the full option universe. The SVCJ implementation is labelled as a proxy because a full affine SVCJ calibration normally requires a time series of option surfaces; here we use a moment-matched variance state so it can be compared fairly with the available live snapshot.

Table 4: Candidate model pricing errors on the full cleaned option universe

Currency	Model	Options	Bias	MAE	RMSE
BTC	MR-ISVM surface	700	19.84	278.15	461.28
BTC	Merton	700	-56.30	328.64	485.53
BTC	Dynamic jump	700	-133.00	349.69	527.01
BTC	GARCH variance	700	-744.28	785.49	1121.61
BTC	Black-Scholes	700	-1297.81	1339.89	1997.38
BTC	SVCJ proxy	700	1439.79	1561.81	2314.15
ETH	MR-ISVM surface	482	-0.47	13.90	23.57
ETH	Merton	482	-12.02	15.98	25.52
ETH	Dynamic jump	482	-12.61	16.43	26.21
ETH	GARCH variance	482	-18.40	19.26	30.81
ETH	Black-Scholes	482	-39.21	39.76	60.64
ETH	SVCJ proxy	482	44.67	48.73	73.70

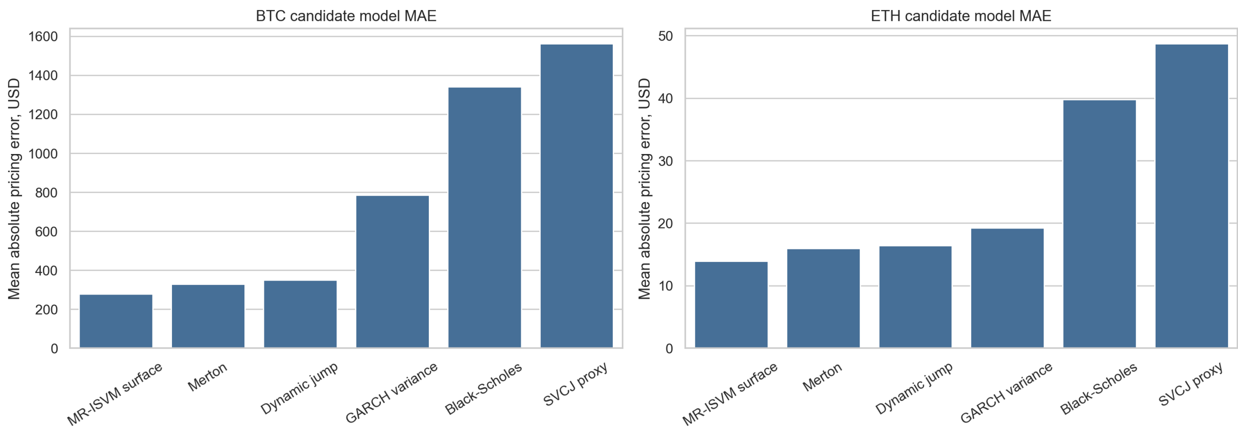

Figure 3: Equal-level candidate comparison using mean absolute pricing error.

Candidate-fit inference. The implied-surface candidate is the strongest cross-sectional fitter because it directly learns the market's smile shape. Merton remains the best structural jump benchmark: it substantially improves on Black-Scholes while retaining a clear risk-neutral jump interpretation. Dynamic jump intensity is competitive but does not dominate Merton in this snapshot once the current intensity is constrained to be at least the base intensity. GARCH explains part of the smile through volatility clustering, but it lacks an explicit discontinuity channel. The SVCJ proxy overprices the option universe because the moment-matched variance-jump state is deliberately conservative without a full time series of option surfaces to discipline variance-jump risk premia.

Table 5: Standardized 30-day 5% OTM put values across candidates

Currency	Model	Put value	Premium over BS
BTC	Black-Scholes	949.19	0.00
BTC	Merton	2146.68	1197.49
BTC	SVCJ proxy	2411.96	1462.77
BTC	MR-ISVM surface	1791.73	842.54
BTC	Dynamic jump	2074.67	1125.48
BTC	GARCH variance	1250.35	301.16
ETH	Black-Scholes	64.19	0.00
ETH	Merton	91.48	27.29
ETH	SVCJ proxy	103.22	39.03
ETH	MR-ISVM surface	77.26	13.07
ETH	Dynamic jump	90.65	26.46
ETH	GARCH variance	80.73	16.54

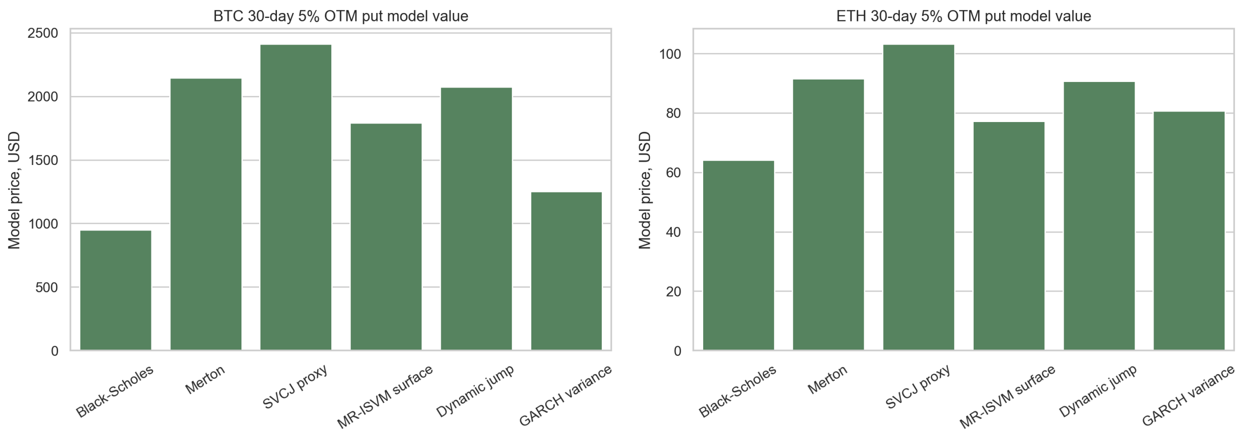

Figure 4: Model-implied crash-protection values for standardized 30-day 5% OTM puts.

Tail-risk inference. The ranking changes when the object is crash protection rather than average pricing error. SVCJ produces the largest 30-day downside protection value because it prices the joint event "spot falls and volatility jumps." Merton and dynamic jump intensity also attach large value to discontinuous downside moves. GARCH is more conservative because volatility clustering alone does not create jump losses. MR-ISVM sits between these cases: it reads the current market smile, but it does not explicitly simulate jump arrival.

5 Historical Distribution and Jump Diagnostics

We compute one-year BTC/ETH log returns from Deribit index history. Jumps are detected using a robust z-score threshold of 4.0. Non-jump observations initialize diffusive volatility; jump observations initialize jump intensity and jump-size distribution.

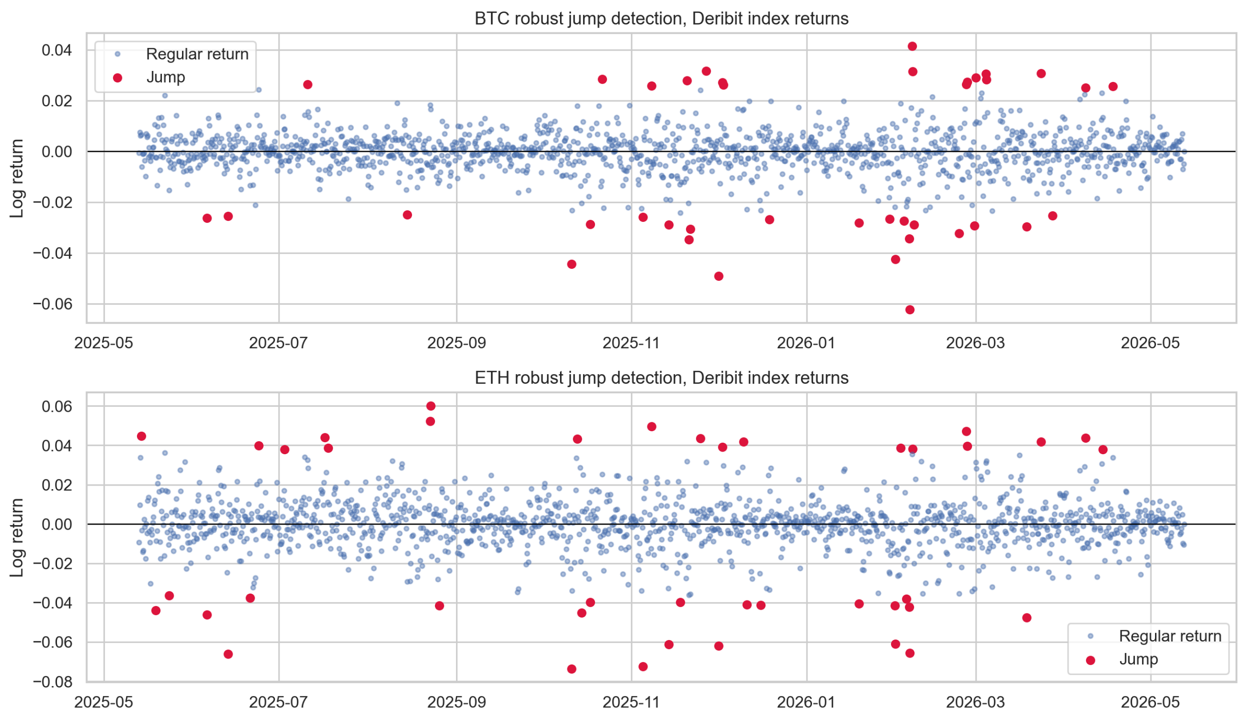

Figure 5: Robust jump detection in BTC and ETH index log returns.

Table 6: Historical jump initialization

Currency	Obs.	Jumps	Diff. σ	λ	μ_J	δ_J	Worst	Best
BTC	1459	39	27.57%	39.05	-0.575%	3.162%	-6.240%	4.139%
ETH	1459	41	44.51%	41.06	-0.640%	4.771%	-7.373%	6.014%

Table 7: Return distribution diagnostics

Currency	Ann. mean	Ann. vol	Skew	Ex. kurtosis	JB statistic	JB p-value
BTC	-23.59%	33.66%	-0.556	4.936	1543.60	$< 10^{-12}$
ETH	-7.58%	53.43%	-0.381	3.496	771.33	$< 10^{-12}$

Inference. Jarque-Bera tests strongly reject normality for both BTC and ETH returns. Both assets exhibit negative skew and excess kurtosis, validating the modeling decision to go beyond a lognormal diffusion. ETH has higher annualized volatility and jump-size volatility, while BTC still exhibits material discontinuous downside risk.

This section deepens the Black-Scholes estimation chapter. In the classical model, the statistic we need is the standard deviation of log returns. For BTC/ETH, that statistic is still useful, but it is incomplete: skewness, kurtosis, and jump frequency explain why a single σ cannot reproduce the market smile. The empirical distribution therefore motivates a pricing model with both a continuous volatility channel and a discontinuous jump channel.

6 Market Calibration and Statistical Evidence

Merton parameters are calibrated to liquid Deribit option marks by minimizing weighted relative pricing error. The calibration uses 80 liquid options per asset; diagnostics are computed on the full cleaned universe. Conceptually, this is an implied-parameter exercise: just as the course defines implied volatility by inverting Black-Scholes, we infer jump parameters by matching a cross-section of market option prices.

Table 8: Black-Scholes versus calibrated Merton pricing error

Currency	Cal. σ	Cal. λ	Cal. μ_J	Cal. δ_J	BS MAE	Merton MAE
BTC	30.67%	39.05	-1.978%	4.356%	1339.89	328.64
ETH	38.88%	41.06	-2.746%	5.726%	39.76	15.98

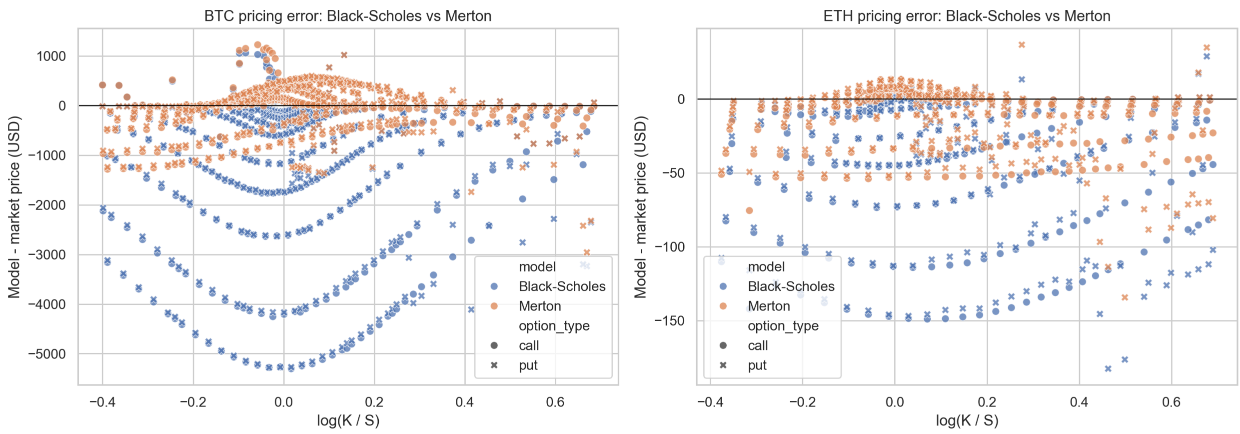

Figure 6: Model pricing errors against Deribit option marks.

Table 9: Paired statistical tests for absolute-error reduction

Currency	Pairs	Mean impr.	Median impr.	Share impr.	t-test p	Wilcoxon p
BTC	700	1011.25	275.91	90.57%	4.79×10^{-71}	6.00×10^{-102}
ETH	482	23.77	7.71	79.88%	7.39×10^{-52}	4.32×10^{-45}

Inference. The Merton improvement is not merely visual. Paired t-tests, Wilcoxon signed-rank tests, and sign tests all show overwhelming evidence that the jump-diffusion model reduces absolute pricing error relative to the Black-Scholes historical-volatility baseline. This is the central empirical support for the innovation.

The paired design is important. Each option is priced twice, once by Black-Scholes and once by Merton, so the test compares models instrument by instrument rather than comparing two unrelated samples. This mirrors the course's emphasis on using a benchmark model first, then measuring model error against market prices.

7 Interest Rate Sensitivity

We price 30-day at-the-money calls and puts under $r = 1\%$ and $r = 5\%$ to isolate interest-rate sensitivity.

Table 10: Interest-rate sensitivity, 30-day at-the-money options

Currency	Option	Rate	BS price	Merton price
BTC	call	1%	2574.10	3926.88
BTC	call	5%	2705.39	4058.36
BTC	put	1%	2507.88	3860.66
BTC	put	5%	2374.84	3727.81
ETH	call	1%	116.46	144.55
ETH	call	5%	120.05	148.20
ETH	put	1%	114.60	142.69
ETH	put	5%	110.74	138.88

Inference. The rate effect has the expected sign from put-call parity and discounting: higher rates increase calls and reduce puts. Over a 30-day crypto horizon, however, jump and volatility assumptions dominate rate sensitivity. This is a useful hierarchy of risks: for short-dated BTC/ETH options, the main modeling battle is not the risk-free rate, but the distribution of the underlying.

8 Jump-Parameter Sensitivity

We stress the calibrated Merton model by repricing 30-day, 5% out-of-the-money puts under different jump-intensity and jump-volatility multipliers. This directly measures the value of crash protection.

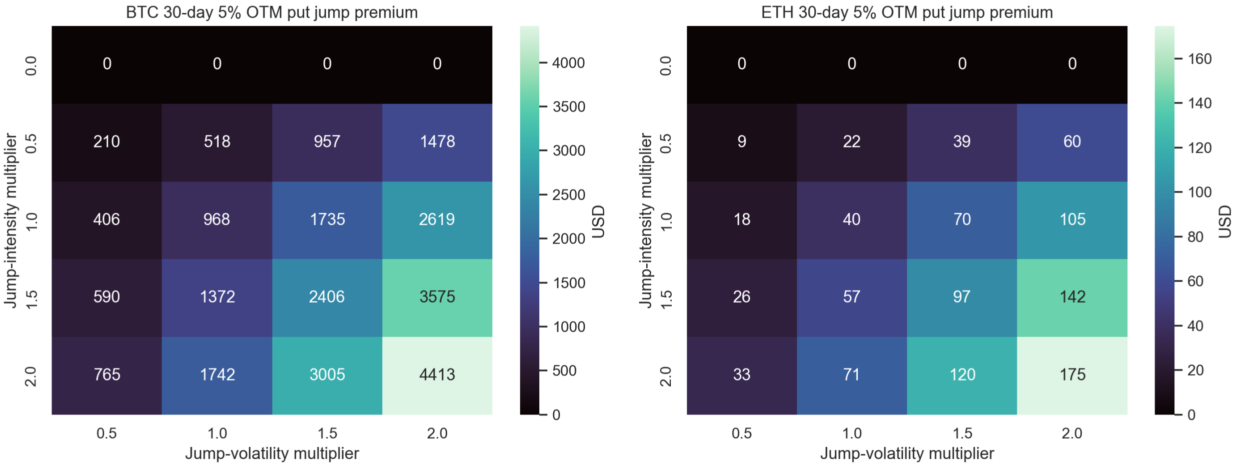

Figure 7: Sensitivity of 30-day 5% OTM put jump premium to jump intensity and jump-size volatility.

Inference. The jump premium rises nonlinearly as either jump intensity or jump-size volatility increases. For BTC, doubling both calibrated jump intensity and jump volatility increases the 30-day 5% OTM put premium by about USD 4,413 relative to the diffusion-only baseline. For ETH, the comparable premium is about USD 175. This is a compact risk-management result: crash protection is primarily a jump-risk instrument, not just a high-volatility instrument.

The Risk Management chapter brings this sensitivity analysis into model-risk language. Greeks measure local sensitivities, but a jump parameter shock is a structural stress test. The heatmap therefore complements Delta and Vega: it asks how much the option value changes when the assumed tail process itself becomes more severe.

9 Greeks and Risk Management

The course handouts define Greeks as sensitivities of option value to state variables and parameters. We compute Black-Scholes Greeks analytically and Merton Greeks by finite differences.

Table 11: Greeks for 30-day at-the-money options at $r = 1\%$

Currency	Option	BS Δ	Merton Δ	BS Γ	Merton Γ	BS Vega	Merton Vega
BTC	call	0.5199	0.5402	0.0000625	0.0000412	9210.15	6753.37
BTC	put	-0.4801	-0.4598	0.0000625	0.0000412	9210.15	6753.37
ETH	call	0.5280	0.5486	0.0013728	0.0011154	259.21	183.99
ETH	put	-0.4720	-0.4514	0.0013728	0.0011154	259.21	183.99

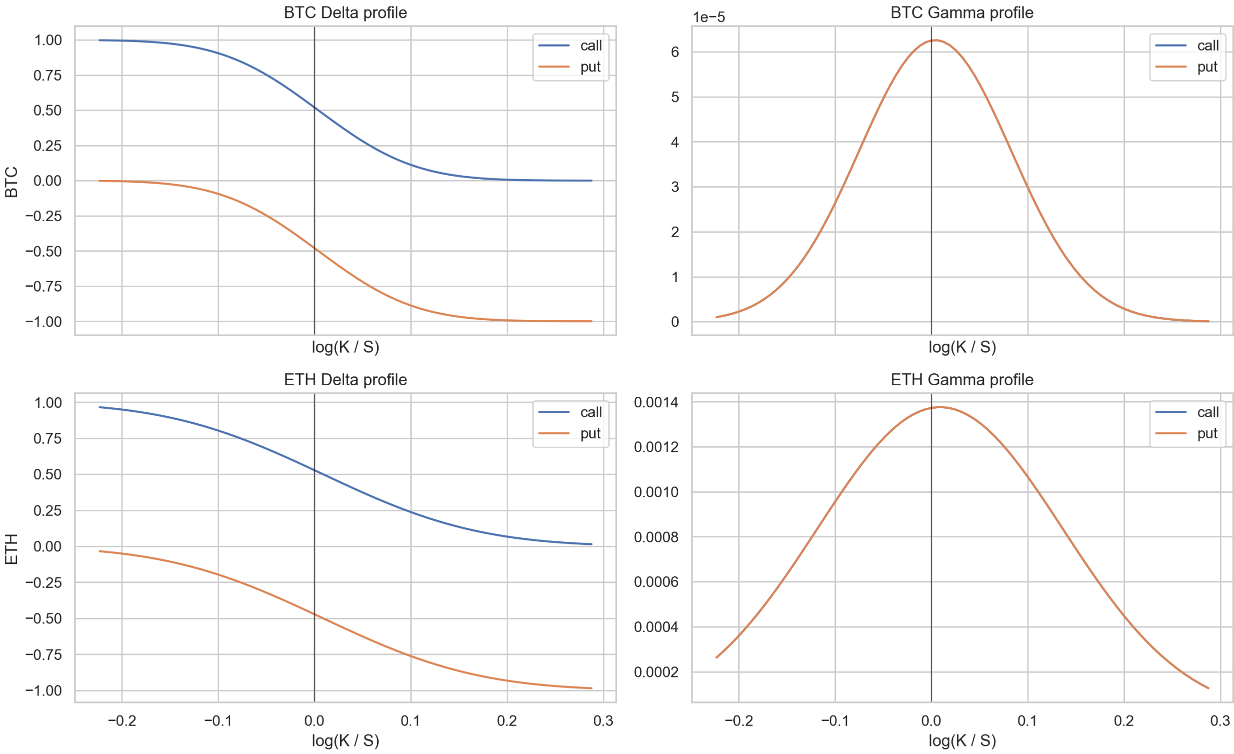

Figure 8: Black-Scholes Delta and Gamma profiles across log-moneyness.

Inference. Merton deltas differ from Black-Scholes deltas because downside jump risk shifts the risk-neutral distribution. Gamma and Vega are also reallocated: some tail value is explained by jump parameters rather than purely by diffusive volatility. A desk hedging only Black-Scholes delta would therefore remain exposed to jump convexity.

The course relation

$$\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = r\Pi$$

is a useful interpretation device. Under pure diffusion, Delta and Gamma summarize the first two local spot risks. Under jumps, however, a finite move in S can dominate the Taylor approximation. This is why the Greeks section must be read together with the jump-sensitivity and hedge-simulation sections.

10 Jump-Hedging Simulation

Continuous delta hedging is an idealization. The Risk Management chapter notes that periodic hedging introduces replication error. We simulate 30-day, 5% OTM put hedging under Merton jump paths.

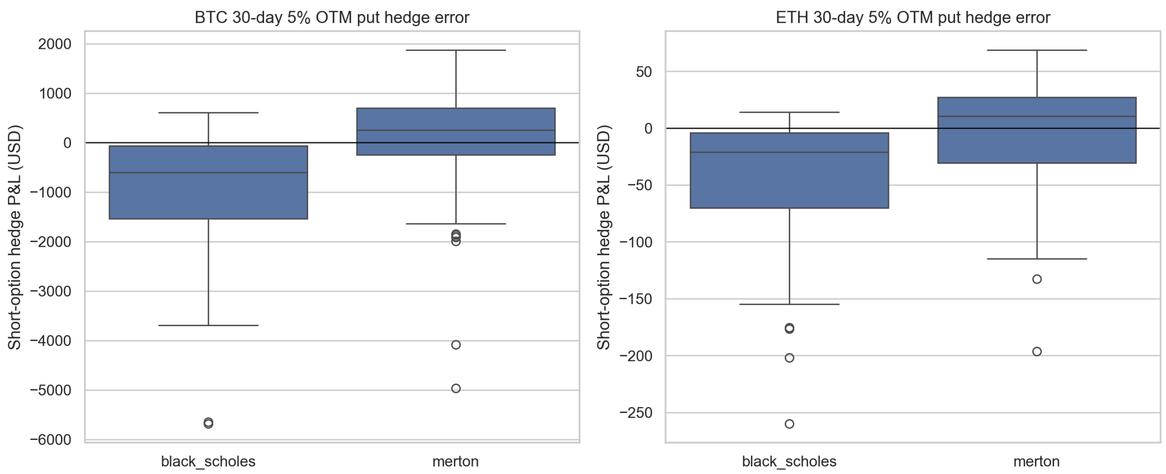

Figure 9: Discrete hedge P&L under Merton jump paths.

Table 12: Discrete hedge-error summary

Currency	Model	Mean P&L	Median P&L	Std P&L	Mean |P&L|	5% P&L
BTC	Black-Scholes	-934.20	-597.65	1188.41	1009.91	-3065.58
BTC	Merton	49.08	251.08	1051.25	739.08	-1869.20
ETH	Black-Scholes	-41.63	-21.24	52.05	43.81	-138.99
ETH	Merton	-3.00	10.44	46.23	34.64	-91.64

Inference. Merton-aware hedging reduces mean absolute hedge error for both assets and materially reduces the BTC left-tail hedge loss. It is not perfect: jumps cannot be hedged after they arrive. But it is more honest about the risk process than a diffusion-only hedge.

This result connects directly to the replication logic in the binomial and Black-Scholes chapters. Replication is exact only under the assumed dynamics and rebalancing frequency. Once the true path contains discontinuities, a self-financing delta hedge accumulates residual error. The practical question is therefore not whether hedging eliminates risk, but which model leaves the smaller residual under realistic crypto paths.

11 Discussion

Black-Scholes is internally coherent under frictionless trading, continuous hedging, constant volatility, and lognormal diffusion. Crypto markets violate these assumptions in several ways:

- returns are heavy-tailed and negatively skewed;
- volatility is regime-dependent;
- investors pay for downside crash protection;
- delta hedging is discrete and liquidity can vanish during stress;
- leverage and liquidation effects can amplify short-horizon moves.

The Merton model is therefore not an ornamental extension. It is a mathematically controlled way to carry the course's risk-neutral framework into the empirical reality of BTC/ETH options. But the equal-level candidate comparison also shows why Merton should not be the final word: MR-ISVM is better at matching the observed option surface, SVCJ is more explicit about volatility jumps, dynamic intensity targets jump clustering, and GARCH isolates volatility persistence as a competing explanation.

The sequence of chapters brings the analysis to a layered conclusion: risk-neutral pricing tells us how to value claims once dynamics are specified; binomial trees show how replication works discretely; Black-Scholes gives the diffusion benchmark; implied volatility reveals the benchmark's market misfit; and risk management exposes the hedging cost of that misfit. The model-comparison layer is our response to that chain of evidence.

12 Conclusion

This project implements a complete option pricing workflow: Black-Scholes prices and Greeks, CRR European and American trees, implied-volatility smiles, market calibration, candidate model comparison, and dynamic hedge simulation.

The innovation is no longer only the Merton jump-diffusion layer. It is the equal-level comparison of five pricing views: diffusion, constant jump intensity, stochastic volatility jumps, regime implied volatility, dynamic jump intensity, and GARCH conditional variance.

The central finding is that crypto option markets price jump risk and volatility-regime risk explicitly. Black-Scholes should be treated as a control model and teaching benchmark. A professional crypto options desk would use a surface model for marking and a structural tail-risk model for stress testing, Greeks, and risk management.

References

[1] Bates, D. S. (1996). Jumps and stochastic volatility: Exchange rate processes implicit in Deutsche Mark options. *The Review of Financial Studies*, 9(1), 69–107.

[2] Black, F., and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.

[3] Chen, K. S., and Yang, J. J. (2024). Price dynamics and volatility jumps in bitcoin options. *Financial Innovation*, 10, Article 132.

[4] Cox, J. C., Ross, S. A., and Rubinstein, M. (1979). Option pricing: A simplified approach. *Journal of Financial Economics*, 7(3), 229–263.

[5] Duffie, D., Pan, J., and Singleton, K. (2000). Transform analysis and asset pricing for affine jump-diffusions. *Econometrica*, 68(6), 1343–1376.

[6] Hou, A. J., Wang, W. N., Chen, C. Y.-H., and Härdle, W. K. (2020). Pricing cryptocurrency options. *Journal of Financial Econometrics*, 18(2), 250–279.

[7] Jiang, W. (2026). IEDA4331 Financial Engineering lecture notes: risk-neutral pricing, binomial trees, Black-Scholes-Merton, and risk management. Hong Kong University of Science and Technology.

[8] Merton, R. C. (1973). Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1), 141–183.

[9] Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.

[10] Saef, D., Wang, Y., and Aste, T. (2022). Regime-based implied stochastic volatility model for crypto option pricing. arXiv:2208.12614.

[11] Venter, P. J., and Maré, E. (2021). Univariate and multivariate GARCH models applied to Bitcoin futures option pricing. *Journal of Risk and Financial Management*, 14(6), Article 261.

[12] Zulfiqar, N., and Gulzar, S. (2021). Implied volatility estimation of Bitcoin options and the stylized facts of option pricing. *Financial Innovation*, 7, Article 67.

A Implementation Notes

The implementation is modular:

- `src/data_deribit.py`: public Deribit data access and caching.
- `src/preprocessing.py`: quote cleaning, USD conversion, returns, and jump detection.
- `src/black_scholes.py`, `src/binomial.py`, and `src/merton_jump.py`: baseline and jump pricing engines.
- `src/candidate_models.py`: comparable modern candidates.
- `src/calibration.py` and `src/risk.py`: calibration and hedge simulation.
- `scripts/generate_report_assets.py`: reproducible report tables and figures.

B AI Assistance Statement

AI assistance was used to structure the project, implement Python modules, draft explanatory text, generate plots and tables, and format this report. The quantitative formulas, course alignment, market data, statistical tests, and validation checks are explicit in the codebase and report. The implementation was verified with unit tests and live Deribit data smoke tests.